

Phased array will be the future of ultrasonic non-destructive testing. Although this technology has been spoken about for quite some time now, it is likely to gain traction in the near future and is therefore worth a watch

ing the various radios sequentially. New systems, on the other hand, will test all these in parallel, reducing the time taken to a fourth of the serial technique's.

Communications testing: Gigabit Ethernet and beyond

Speaking about the assessment of next-generation networks (NGN), gigabit Ethernet has taken the T&M world by storm. Test products for Past-10 gigabit networks are already being demonstrated.

At the *Optical Fibre Communication Conference and Exposition and The National Fibre Optic Engineers Conference (OFC/NFOEC 2010)*, several such products were showcased, and experts even opined that 100 gigabit Ethernet will arrive in core networks this year.

Some companies demonstrated optical modulation analysers capable of 244Gbps data streams, while others showcased 10G Ethernet testers with electrical and optical ports for 10Mbps through 10Gbps links. Others products included 12.5Gbps parallel-channel BER testers, 40Gbps and 100Gbps BER testers, clock-recovery modules and clock synthesisers.

With so many greater-than-10G products on the market, it is time to think about co-existence. Not surprisingly, in March this year, several network test-equipment manufacturers, networking equipment manufacturers and service providers came together to form the Network Test Automation Forum, to define an open and advanced automation framework. The idea is to develop a software layer that resides between network equipment such as switches and routers and network test

Innovative Products Launched Last Year...

Amp Tronics 3-in-1 Handheld

Multiple equipment packaged in a single handheld device is something very useful for field engineers. The HDS1021M from Amp Tronics can perform 18 automatic measurements with DSO, multimeter and cymometer packaged in a low-cost handheld device. The combination of DSP and DMM highly improves the test efficiency and convenience. It also features a TFT true-colour display and supports USB for data transmission. The Li-ion battery provides backup for up to six hours.

This 3-in-1 handheld is designed using ARM9 development board with a high-performance microprocessor. (For details, visit www.amp-tronics.com)



ArbStudio Waveform Generator From Lecroy

ArbStudio waveform generator from Lecroy provides 125MHz bandwidth, 16-bit resolution, two- and four-channel options, 1GS/s maximum sampling rate and memory depth of 2 Mpts per channel. It generates analogue and digital patterns with 18 and 36 channels, respectively. An intuitive interface helps you to create, edit and sequence waveforms. ArbStudio provides easy-to-use tools with built-in modulation capabilities like amplitude-shift keying (ASK), pulse-shift keying (PSK) and frequency-shift keying (FSK) that can be configured for any application.

ArbStudio is designed using a high-performance FPGA. Use of FPGA enables both arbitrary as well as digital synthesis modes in a single equipment, thereby reducing the price considerably. Some of Math and noise functions are built-in and can be combined with the waveforms. (Distributed in India by Electro Systems Associates (www.esaindia.com))



Kusam DMM for VFD Measurements

A new digital multimeter from Kusam Electrical Industries, DMM 869, offers a low-cost solution to almost 39 different measurements for any lab. Apart from the regular multimeter functions, it incorporates variable-frequency drive (VFD) measurements. In automation industries, for motor drive applications, VFD is used for power and energy savings. VFD measurements and VFD high-frequency-band voltage (100kHz) measurements help in ensuring the same. The device also provides surge protection up to 12 kV, CAT4 protection up to 600V and CAT3 protection up to 1000V.

In order to design a high-end DMM with a variety of functions, a custom-made microprocessor technology has been used. (For details, visit www.kusamelectrical.com/)

